PAPER_610: Mayan Calendar Cosmological Epochs Mapped to Periodic Table Nuclei Formation

Unified Quantum Field Framework (UQFF)

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Abstract

The Mayan calendar's five cosmological epochs (Baktun cycles) are mapped to five phases of nuclei formation in the UQFF framework via 3D-IPO (symbolic + numerical + discrete) convergences. Prime atomic numbers anchor stable elements at epoch transitions. Z=1 (hydrogen) emerges in epoch 1; helium through beryllium in epoch 2; carbon through zinc in epoch 3; heavy elements (Z=31–118) in epoch 4; and speculative superheavy stable islands (Z>118) in epoch 5. This provides a cyclical cosmological account of the periodic table.

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1. Introduction: Cycles and Elements

The Maya Baktun cycle (~394 years) encodes cosmological time periods. In UQFF, each Baktun epoch corresponds to a convergence of DPM grinding energy with the SCm injection layers, producing successive groups of stable nuclei. The 3D-IPO method simultaneously solves the system symbolically (pyramid arithmetic), numerically (Orion parameters), and discretely (Wolfram hypergraph rules).

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2. Five Epochs and Their Z-Ranges

Epoch	Mayan Units	UQFF Phase	Z Range	Representative Elements
1	1st Great Cycle	Proto-Universe	Z=1	H (from 26D shell alignment)
2	2nd Great Cycle	First Stars	Z=2–4	He, Li, Be
3	3rd Great Cycle	Galactic Nucleosyn.	Z=5–30	B,C,N,O,F,Ne,...Zn
4	4th Great Cycle	Supergalactic	Z=31–118	Ga...Og (all known)
5	5th Great Cycle (future)	Post-cosmic	Z>118	Island of stability (Z=120,126?)

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3. 3D-IPO Convergence Method

Each epoch forms nuclei through the simultaneous satisfaction of three constraints:

Symbolic (pyramid arithmetic): $$Z_{stable} = \sum_{n=1}^{epoch} pyramid_n \pmod{\text{shell-closure rules}}$$ where $pyramid_n = n(n+1)/2$ gives triangular numbers matching magic nuclear numbers (2, 8, 20, 28, 50, 82, 126...).

Numerical (Orion parameters): $$E_{epoch} = h \cdot f_{Orion} \cdot epoch = 6.626\times10^{-34} \times 6.93\times10^9 \times epoch$$

Epoch	E_epoch (J)
1	$4.59\times10^{\blacksquare 2\blacksquare}$
2	$9.18\times10^{\blacksquare 2\blacksquare}$
3	$1.38\times10^{\blacksquare 23}$
4	$1.84\times10^{\blacksquare 23}$
5	$2.30\times10^{\blacksquare 23}$

Discrete (Wolfram hypergraph): Each epoch corresponds to one Wolfram rule application in the hypergraph node-rewriting system. The discrete transitions produce unique nuclear fingerprints consistent with observed isotope abundances.

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4. Prime Z-Anchors (DVP Connection)

At epoch transitions, the first new element is always a DVP prime:

- Epoch 1 → Z=1 (not prime, but protonic)
- Epoch 2 → Z=2 (He, prime)
- Epoch 3 → Z=5 (B, prime)
- Epoch 4 → Z=31 (Ga, prime)
- Epoch 5 → Z=?, predicted next prime ≥ 119 = 7×17 → Z=127 (prime, predicted island)

The pattern suggests DVP prime-indexed elements are the most stable at each epoch boundary, consistent with nuclear magic numbers and known islands of stability (Z=82,126 are near primes).

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5. Speculative Fifth Epoch: Z > 118

The heaviest known element is Oganesson (Z=118, Og). UQFF predicts that in the 5th epoch, DPM grinding at the highest shell layer creates nuclei beyond Z=118 with half-lives measured in years or longer. The DVP prime Z=127 is predicted to anchor the new island of stability, corresponding to the 3rd shell closure beyond Z=118.

This is experimentally testable at GSI/Darmstadt, RIKEN, and JINR.

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6. Connection to Mayan Calendar Cosmology

The Mayan Long Count calendar's end of the 13th Baktun (Dec 21, 2012) in UQFF corresponds to the transition from epoch 4 to epoch 5 — from known elements to the speculative superheavy island. This is not mysticism but an encoding of nuclear physics timescales: each Baktun (1,872,000 days ≈ 5,125 years) scales to a cosmological nucleosynthesis phase.

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7. Connection to UQFF Number Systems

DVP: Primes Z=2,3,5,7,11,... are DVP nuclear anchors. Each prime Z corresponds to one DVP vortex prime-indexed orbital shell. **VDS**: Shell energies per epoch follow VDS expansion: $E_{shell}(Z) \propto \sum d_n(\pi)/Z^n$. **BH26**: The 5 epochs correspond to layers 1–5 of the 26 BH26 harmonic bins most relevant for nucleosynthesis; the remaining 21 are cosmological background.

Keywords: Mayan calendar, periodic table, nuclei epochs, 3D-IPO convergence, DVP, prime Z, island of stability, UQFF

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